

```
In[10]:= dxold = 0.4; dyold = 0.08; xi = 2.9; yi = 4.65;
```

```
z[x_, y_] := 3 * Cos[ $\pi$  * x / y];
```

```
dxnew = dxold *  $\sqrt{\frac{10}{40}}$ ; dynew = dyold *  $\sqrt{\frac{15}{240}}$ ;
```

```
z[xi, yi]
```

```
Out[13]=
```

```
-1.13634
```

```
In[28]:= Sqrt[(D[z[x, y], x] * dxnew)2 + (D[z[x, y], y] * dynew)2] /. {x -> xi, y -> yi}
```

```
Out[28]=
```

```
0.375891
```